



[BIT703 Web Technologies](#) / [Contents](#) / Topic 7: Client-side web application development and programming – JavaScript / UI frameworks for client-side development

# Introduction to frameworks 7m

Many developers who work with JavaScript rarely use JavaScript alone. Rather, they use libraries and frameworks that simplify much of the complexity of the underlying language. JavaScript libraries and frameworks are both JavaScript code, but they differ in the way they are used.

A framework acts as a template or scaffold for the code, defining behaviour, architecture, functionalities and flow control of a web application. Frameworks provide many integrated underlying services, like templating and data management, that make it easy to implement the coding aspects that frameworks support. But unlike libraries, frameworks force code to be written in a specific way, which means they can become core to how the web application will work, making them harder to replace or substitute.

According to the 2021 Stack Overflow developer survey, React.js was the most popular client-side JavaScript web framework (Stack Overflow, 2021).

## Benefits of using frameworks

Here are some of the benefits of using frameworks:

- You don't have to reinvent code that could well be buggy and unstable.
- There is additional functionality over the inbuilt features of the language.
- There is support from the user community of other developers.
- Integration and interoperability with other systems are provided.
- You can focus on designing and developing with larger-grained components rather than writing raw code.

[Skip to main content](#)

When selecting a framework, it is always best practice to determine the level of support and documentation available. A framework could be a perfect solution regarding reinventing code but might prove difficult to use owing to a lack of suitable documentation and support.

It's worthwhile to check whether someone has published books about a particular framework. React and AngularJS are the subject of many books, which makes them good options to start with – though, of course, these books can fall out of date quickly as frameworks evolve.

Once a suitable framework has been identified, the developer needs to determine how to integrate it into their application. Many third-party frameworks have different installation procedures and methods. Therefore, it's not always a simple include-and-use approach.

In this section, we will discuss the JavaScript frameworks React, AngularJS and Vue.js.

## Dig deeper

To learn more about the basics of JavaScript frameworks, read these resources:

- *Introduction to Client-Side Frameworks* (MDN, 2025) – this article gives an overview of what frameworks are and why they're used, as well as how to choose a framework.
- *Framework Main Features* (MDN, 2025) – this article gives an overview of the major JavaScript frameworks' main features and how these frameworks work at a high level.



[Skip to main content](#)

Thoughts on this page?

